

Sazadur Rahman

 Scholarly
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About Me

I am a tenure-track Assistant Professor at the Department of Electrical and Computer Engineering at University of Central Florida (UCF). At UCF, I lead a group of 6 PhD students. Our lab focuses on hardware security, AI-based electronic design automation, secure packaging, and photonic integrated circuit. We regularly tape-out our developed solutions with recent experience in 14nm, 28nm, 65nm, and 90nm technologies from TSMC, MIT-LL, and Tiny-Tapeout. I developed and teach the commercial chip design and tapeout class at UCF (Spring'25,'26) with next tapeout scheduled for Dec'26.

Education

University of Florida *Jan 2018 – Dec 2022*

Ph.D. and M.Sc. in Computer Engineering

- **Supervisor:** Prof. Mark Tehranipoor, Chair, Department of ECE, UF
- **Research:** Hardware Security, Intellectual Property Protection and Authentication, Side-channel Analysis, Artificial Intelligence based Security Metrics, and Secure Development Lifecycle.

Bangladesh University of Engineering and Technology (BUET) *Mar 2009 – July 2014*

B.Sc. in Electrical and Electronic Engineering

- **Supervisor:** Prof. Nasim Ahmed Dewan
- **Research:** Ion Energy Distribution of Multi-frequency Capacitively Plasma

Experience

Assistant Professor, University of Central Florida *Orlando, FL*
Cyber Security and Privacy Cluster *Aug 2023 – Present*

Department of Electrical and Computer Engineering
CREOL, The College of Optics and Photonics (Joint)

- **Research:** Hardware Security, AI-based Electronic Design Automation, Heterogeneous Integration, Trustworthy AI, and Confidential Computing.
- **Teaching:** Computer architecture (Fall'23, Fall'24), Full Custom VLSI Design (Spring'25).

Security Architecture Engineer *Gainesville, FL*

Intel Product Assurance and Security (IPAS) Team, Intel Corporation *Jan 2023 – Aug 2023*

- Worked for architectural security hardening of Intel IPs against unauthorized access, microarchitectural, telemetry, and power side-channel vulnerabilities.

Security Researcher Intern *Gainesville, FL*

Intel Product Assurance and Security (IPAS) Team, Intel Corporation *May 2021 – Aug 2021*

- Developed automated threat modeling review tool utilizing CWE, CVE, ATT&CK, and CAPEC databases from mitre.org for different adversary models based on their capabilities.

Hardware Security Intern *Hillsboro, OR*

Intel Memory Security Team, Intel Corporation *Jan 2020 – May 2020*

- Developed firmware for FIPS 140 -3 security certification of cryptographic hardware using NIST Cryptographic Algo. Verification Program (CAVP).

Senior Design Engineer *Dhaka, Bangladesh*

Neural Semiconductor Ltd., and PrimeSilicon Technologies Ltd. *Apr 2017 – Dec 2017*

- Lead a team of 20 engineers for custom ASIC design.

Senior ASIC Physical Design Engineer *Dhaka, Bangladesh*

PrimeSilicon Technologies Ltd. *Sept 2014 – Mar 2017*

- Taped out two 14nm 2.5D FinFET network processor (35M & 50M gates) and two 28nm network controller (57M & 140M gates) using Samsung 14nm LPP PDK, HBM2PHY & Rambus 28G SerDes.

Teaching

Classes Taught

Semester	Course Number	Course Title	Student Count	SPI
Spring 2026	EEE5390C	Full Custom VLSI Design	29	4.5
Fall 2025	EEL4334/5336	Introduction to VLSI	33	4.35
Fall 2025	EEL4768	Computer Architecture	50	4.31
Spring 2025	EEE5390C	Full Custom VLSI Design	30	4.40
Fall 2024	EEL4768	Computer Architecture	48	3.64
Fall 2023	EEL4768	Computer Architecture	55	4.29

PhD Students Supervised

1. Ishraq Tashdid (Spring'24 -)
2. Kimia Tasnia (Spring'24 -)
3. Dewan Saiham (Summer'24 -)
4. Tanveerul Islam (Fall'25 -)
5. Ahmed Mustakim (Spring'26 -)
6. Md Mehedi Hasan (Fall'26 -)

Undergraduate Students Supervised

1. Jacob Lingo (Spring'24 - Spring'25) - Now at Mitsubishi Power
2. Valentina Terry (Spring'24 - Fall'25) - Now at Apple
3. Alexander Garcia (Summer'24 - Spring'26) - Now at AMD
4. Krystal Kelly (Summer'24 - Summer'25)
5. Hilda Deveaux (Fall'24 - Spring'26)
6. Jordan Merkel (Fall'24 - Summer'25)
7. Imani-Hans Ngalame (Fall'24 - Fall'25)
8. John Gierlach (Fall'23 - Fall'25) - Now at AMD
9. Mariam Rabadi (Fall'23 - Fall'24) - Now at Apple

Publications

With My Students.

1. Ishraq Tashdid, Kimia Tasnia, Alexander Garcia, Jonathan Valamehr, and **Rahman, Sazadur**. Atlas: Ai-assisted threat-to-assertion learning for system-on-chip security verification. *2026 IEEE/ACM Design Automation Conference (DAC)* <https://arxiv.org/abs/2603.01170>, 2026
2. Dewan Saiham and **Rahman, Sazadur**. Workload-induced thermal impact and dvfs control in co-packaged photonic interconnects. *IEEE Research and Applications of Photonics in Defense Conference (RAPID)*, 2026

3. Dewan Saiham, Md Omar Faruk Noman, **Rahman, Sazadur**, and Soumitra Joy. Mitigating clock glitch attack via electromagnetically engineered interconnects. In *ACM/IEEE International Symposium on Low Power Electronics and Design*, pages 1–7, 2026
4. Dewan Saiham, Di Wu, and **Rahman, Sazadur**. Optolink: Breaking memory bandwidth bottlenecks in the accelerators with photonic interconnects. *ACM Transactions on Design Automation of Electronic Systems*, 2026 <https://dl.acm.org/doi/10.1145/3806395>
5. Ishraq Tashdid, Tasnuva Farheen, and **Sazadur Rahman**. Interpuf: Distributed authentication via physically unclonable functions and multi-party computation for reconfigurable interposers. In *2026 IEEE international symposium on hardware-oriented security and trust (HOST)*. IEEE, 2026 <https://arxiv.org/abs/2601.11368>
6. Dewan Saiham and **Rahman, Sazadur**. Workload-aware co-thermal optimization of photonic placement in co-packaged optics. In *Reconfigurable Photonic Computing and Heterogeneous Architectures for AI and RF Defense Systems*, volume 14024, page 57. SPIE, 2026
7. Ishraq Tashdid, Valentina Terry, Jordan Merkel, Tasnuva Farheen, and **Rahman, Sazadur**. Beyondppa: Human-inspired reinforcement learning for post-route reliability-aware macro placement. In *2025 ACM/IEEE 7th Symposium on Machine Learning for CAD (MLCAD)*, **Best Paper Nominated**, pages 1–12. IEEE, 2025 <https://doi.org/10.1109/MLCAD65511.2025.11189164>
8. Ishraq Tashdid, Tasnuva Farheen, and **Rahman, Sazadur**. Authentree: A scalable mpc-based distributed trust architecture for chiplet-based heterogeneous systems. *2025 IEEE International Conference on PHYSICAL ASSURANCE and INSPECTION of ELECTRONICS (PAINE)*, <https://doi.org/10.1109/PAINE66113.2025.11320196>
9. Ishraq Tashdid, Dewan Saiham, Nafisa Anjum, Tasnuva Farheen, and **Rahman, Sazadur**. Ecologic: Enabling circular, obfuscated, and adaptive logic via efpga-augmented socs. *2025 IEEE International Conference on Computer Design (ICCD)*, <https://doi.org/10.1109/ICCD65941.2025.00080>
10. Kimia Tasnia and **Rahman, Sazadur**. Opl4gpt: An application space exploration of optimal programming language for hardware design by llm. In *Proceedings of the 30th Asia and South Pacific Design Automation Conference*, pages 981–987, 2025 <https://dl.acm.org/doi/abs/10.1145/3658617.3697621>
11. Dewan Saiham, Di Wu, and **Sazadur Rahman**. Leveraging photonic interconnects for scalable and efficient fully homomorphic encryption. In *GOMACTech 2025 - Proceedings in Defense Technical Information Center (DTIC)*., [Arxiv](#) [↗](#)
12. Kimia Tasnia, Alexander Garcia, Tasnuva Farheen, and **Rahman, Sazadur**. Veriopt: Ppa-aware high-quality verilog generation via multi-role llms. *2025 IEEE International Conference on Computer Aided Design (ICCAD)*, <https://doi.org/10.1109/ICCAD66269.2025.11240771>
13. Dewan Saiham, Di Wu, and **Sazadur Rahman**. Can photonic interconnects be used for high-throughput memory access in the accelerators? In *2025 International Symposium on Low Power Electronic Design (ISLPED)*, 2025 <https://doi.org/10.1109/ISLPED65674.2025.11261747>
14. Ishraq Tashdid, Tasnuva Farheen, and **Rahman, Sazadur**. Safe-sip: Secure authentication framework for system-in-package using multi-party computation. In *Proceedings of the Great Lakes Symposium on VLSI 2025*, pages 391–396, 2025 <https://dl.acm.org/doi/full/10.1145/3716368.3735248>
15. Ishraq Tashdid, Tasnuva Farheen, Navid Asadi, and **Rahman, Sazadur**. Routepuf: An interposer-resident delay puf with mpc for zero-trust chiplet sips. In *GOMACTech 2026 - Proceedings in Defense Technical Information Center (DTIC)*, 2026 (Just Accepted)

With my affiliations [Conferences].

1. Cory Brynds, Parker McLeod, Lauren Caccamise, Asmita Pal, Dewan Saiham, **Rahman, Sazadur**, Joshua San Miguel, and Di Wu. Cryptoracle: A modular framework to characterize fully homomorphic encryption. <https://arxiv.org/abs/2510.03565>, 2025
2. Jiaming Wu, Olivia Dizon-Paradis, Sazadur Rahman, Damon Woodard, and Domenic Forte. Doscrack: Deobfuscation using oracle-guided symbolic execution and clustering of binary security keys. In *2024 IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*, pages 227–232. IEEE Computer Society, 2024 <https://doi.org/10.1109/HOST55342.2024.10545388>
3. Bulbul Ahmed, **Sazadur Rahman**, Kimia Zamiri Azar, Farimah Farahmandi, Fahim Rahman, and Mark Tehranipoor. Seamless: Security evaluation of logic locking using machine learning oriented estimation. In *Proceedings of the Great Lakes Symposium on VLSI 2024*, pages 489–494, 2024 <https://doi.org/10.1145/3649476.3660382>
4. Minyan Gao, **Sazadur Rahman**, Nitin Varshney, Mark Tehranipoor, and Domenic Forte. iprobe: internal shielding approach for protecting against front-side and back-side probing attacks. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 42(12):4541–4554, 2023 <https://doi.org/10.1109/TCAD.2023.3276525>
5. Mridha Md Mashahedur Rahman, **Sazadur Rahman**, Rasheed Kibria, Mike Borza, Bandy Reddy, Adam Cron, Fahim Rahman, Mark Tehranipoor, and Farimah Farahmandi. Capec: A cellular automata guided fsm-based ip authentication scheme. In *2023 IEEE 41st VLSI Test Symposium (VTS)*, pages 1–8. IEEE, 2023 <https://doi.org/10.1109/VTS56346.2023.10140093>
6. **Sazadur Rahman**, Nitin Varshney, Farimah Farahmandi, Navid Asadi Zanjani, and Mark Tehranipoor. Lle: mitigating ic piracy and reverse engineering by last level edit. In *International Symposium for Testing and Failure Analysis*, volume 84741, pages 360–369. ASM International, 2023 <https://doi.org/10.31399/asm.cp.istfa2023p0360>
7. **Sazadur Rahman**, Kimia Zamiri Azar, Farimah Farahmandi, and Hadi Mardani Kamali. Metrics-to-methods: Decisive reverse engineering metrics for resilient logic locking. In *Proceedings of the Great Lakes Symposium on VLSI 2023*, pages 685–690, 2023 <https://doi.org/10.1145/3583781.3590273>
8. Upoma Das, **Sazadur Rahman**, N Nalla Anandakumar, Kimia Zamiri Azar, Fahim Rahman, Mark Tehranipoor, and Farimah Farahmandi. Psc-watermark: power side channel based ip watermarking using clock gates. In *2023 IEEE European Test Symposium (ETS)*, pages 1–6. IEEE, 2023 <https://doi.org/10.1109/ETS56758.2023.10174052>
9. Rui Guo, **Sazadur Rahman**, Hadi M Kamali, Fahim Rahman, Farimah Farahmandi, and Mark Tehranipoor. Evolute: evaluation of look-up-table-based fine-grained ip redaction. In *2023 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, **Best Paper Nominated**, pages 1–6. IEEE, 2023 <https://doi.org/10.23919/DATE56975.2023.10137322>
10. Md Saad Ul Haque, Rui Guo, **Sazadur Rahman**, Hadi Mardani Kamali, Farimah Farahmandi, and Mark Tehranipoor. Shi-lock: Enabling co-obfuscation for secure heterogeneous integration against re and cloning. In *2023 IEEE Physical Assurance and Inspection of Electronics (PAINE)*, pages 1–7. IEEE, 2023 <https://doi.org/10.1109/PAINE58317.2023.10317951>
11. Zahin Ibnat, **Sazadur Rahman**, Mridha Mashahedur Rahman, Hadi Mardani Kamali, Mark Tehranipoor, and Farimah Farahmandi. Actiwater: Adaptive and design-agnostic active watermarking for ip ownership in modern socs. In *2023 60th ACM/IEEE Design Automation Conference (DAC)*, pages 1–6. IEEE, 2023 <https://doi.org/10.1109/DAC56929.2023.10247688>
12. M Tanjidur Rahman, Shahin Tajik, **Sazadur Rahman**, Mark Tehranipoor, and Navid Asadizanjani. The key is left under the mat: On the inappropriate security assumption of logic locking schemes. In *2020 IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*, pages 262–272. IEEE, 2020 <https://doi.org/10.1109/HOST45689.2020.9300258>

13. **Sazadur Rahman**, Henian Li, Rui Guo, Fahim Rahman, Farimah Farahmandi, and Mark Tehranipoor. Ll-atpg: Logic-locking aware test using valet keys in an untrusted environment. In *2021 IEEE International Test Conference (ITC)*, pages 180–189. IEEE, 2021 <https://doi.org/10.1109/ITC50571.2021.00026>
14. Rasheed Kibria, **Sazadur Rahman**, Farimah Farahmandi, and Mark Tehranipoor. Rtl-fsmx: Fast and accurate finite state machine extraction at the rtl for security applications. In *2022 IEEE International Test Conference (ITC)*, pages 165–174. IEEE, 2022 <https://doi.org/10.1109/ITC50671.2022.00024>
15. **Sazadur Rahman**, Rui Guo, Hadi M Kamali, Fahim Rahman, Farimah Farahmandi, Mohamed Abdel-Moneum, and Mark Tehranipoor. O'clock: lock the clock via clock-gating for soc ip protection. In *Proceedings of the 59th ACM/IEEE Design Automation Conference*, pages 775–780, 2022 <https://doi.org/10.1145/3489517.3530542>

With my affiliations [Journals].

1. Rui Guo, **Sazadur Rahman**, Jingbo Zho, Hadi M Kamali, Fahim Rahman, Farimah Farahmandi, and Mark Tehranipoor. Evolute+: Fine-grained look-up-table-based rtl ip redaction. In *2025 IEEE Transactions on Computer aided Design (TCAD)*, pages 1–14. IEEE, 2025 <https://eprint.iacr.org/2025/484>
2. Upoma Das, **Rahman, Mohammad**, Akshay Kulkarni, Mark Tehranipoor, and Farimah Farahmandi. Pscmark: Power side channel-based watermarking for soc ips using clock gates. *ACM Transactions on Design Automation of Electronic Systems*, 30(5):1–27, 2025 <https://doi.org/10.1145/3747293>
3. Zahin Ibnat, Mrida Md Mashahedur Rahman, **Rahman, Sazadur**, Jingbo Zhou, and Farimah Farahmandi. Redefining tradition: An active watermarking approach for ip protection in socs. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2025 <https://doi.org/10.1109/TCAD.2025.3611871>
4. Jiaming Wu, Olivia Dizon-Paradis, Sazadur Rahman, Damon L Woodard, and Domenic Forte. Protecting dynamically obfuscated scan chain architecture from doscrack with trivium pseudo-random number generation. *Cryptography*, 9(1):6, 2025 <https://doi.org/10.3390/cryptography9010006>
5. **Sazadur Rahman**, Rui Guo, Hadi M Kamali, Fahim Rahman, Farimah Farahmandi, and Mark Tehranipoor. Retrustedfsm: toward rtl hardware obfuscation-a hybrid fsm approach. *IEEE Access*, 11:19741–19761, 2023 <https://doi.org/10.1109/ACCESS.2023.3244902>
6. N Nalla Anandakumar, **Sazadur Rahman**, Mridha Md Mashahedur Rahman, Rasheed Kibria, Upoma Das, Farimah Farahmandi, Fahim Rahman, and Mark M Tehranipoor. Rethinking watermark: Providing proof of ip ownership in modern socs. *Cryptology ePrint Archive*, 2022 <https://eprint.iacr.org/2022/092>
7. **Sazadur Rahman**, Adib Nahiyani, Fahim Rahman, Saverio Fazzari, Kenneth Plaks, Farimah Farahmandi, Domenic Forte, and Mark Tehranipoor. Security assessment of dynamically obfuscated scan chain against oracle-guided attacks. *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 26(4):1–27, 2021 <https://doi.org/10.1145/3444960>
8. M Tanjidur Rahman, **Sazadur Rahman**, Huanyu Wang, Shahin Tajik, Waleed Khalil, Farimah Farahmandi, Domenic Forte, Navid Asadizanjani, and Mark Tehranipoor. Defense-in-depth: A recipe for logic locking to prevail. *Integration*, 72:39–57, 2020 <https://doi.org/10.1016/j.vlsi.2019.12.007>

Book Chapters

1. Farimah Farahmandi, **Sazadur Rahman**, Sree Ranjani Rajendran, and Mark Tehranipoor. CAD for Hardware Security, 2023 <https://doi.org/10.1007/978-3-031-26896-0>
2. Navid Asadizanjani, Mir Tanjidur Rahman, and Mark Tehranipoor. Physical assurance. *Cham Switzerland: Springer Nature Switzerland AG (Chapter 1)*, 2021 <https://doi.org/10.1007/978-3-030-62609-9>
3. Swarup Bhunia and Mark M Tehranipoor. *Hardware security: a hands-on learning approach*. Morgan Kaufmann (Chapter 6), 2018 <https://doi.org/10.1016/C2016-0-03251-5>

Patents

1. Farimah Farahmandi, Zahin Ibnat, Mark Tehranipoor, **Rahman, Mohammad Sazadur**, Hadi Mardani Kamali, and Mridha Md Mashahedur Rahman. Adaptive and design-agnostic active watermarking for authentication of hardware intellectual property (ip) core ownership, October 30 2025. US Patent App. 19/188,521 <https://patents.google.com/patent/US20250335569A1/en>
2. Mark M Tehranipoor, Domenic J Forte, Farimah Farahmandi, Adib Nahyan, Fahim Rahman, and **Sazadur Rahman**. Protecting obfuscated circuits against attacks that utilize test infrastructures, January 11 2022. US Patent 11,222,098 <https://patents.google.com/patent/US11222098B2/en>
3. Mark M Tehranipoor, Farimah Farahmandi, Hadi Mardani Kamali, **Sazadur Rahman**, Rui Guo, and Fahim Rahman. Clock gating system and method for protecting hardware designs, December 19 2024. US Patent App. 18/734,443 <https://patents.google.com/patent/US20240419879A1/en>
4. Mark M Tehranipoor, **Sazadur Rahman**, Kimia Zamiri Azar, Hadi Mardani Kamali, Fahim Rahman, Farimah Farahmandi, and Rui Guo. Building and redaction of universal functional modules, January 2 2025. US Patent App. 18/745,588 <https://patents.google.com/patent/US20250005241A1/en>
5. Farimah Farahmandi, Zahin Ibnat, Mark Tehranipoor, **Rahman, Mohammad Sazadur**, Hadi Mardani Kamali, and Mridha Md Mashahedur Rahman. Adaptive and design-agnostic active watermarking for authentication of hardware intellectual property (ip) core ownership, October 30 2025. US Patent App. 19/188,521 <https://patents.google.com/patent/US20250335569A1/en>

Honors and Awards

1. 2022 IEEE/ACM DAC PhD Forum Best doctoral dissertation competition finalist
2. 2022 IEEE VTS TTTC's E. J. McCluskey Best Doctoral Thesis Competition – Runner up
3. 2023 IEEE/ACM Design Automation and Test in Europe (DATE) Best Paper Award Nomination (2)
4. 2023 IEEE/ACM DATE PhD Forum Best doctoral dissertation competition finalist
5. 2023-24 UCF ECE AMD Undergraduate Fellowship – Mariam Rabadi, and John Gierlach (undergraduate supervisee)
6. 2024-25 UCF ECE Northrop Grumman Undergraduate Fellowship – Valentina Terry (undergraduate supervisee)
7. 2024-25 UCF ECE Lockheed Martin Undergraduate Fellowship – Hilda Deveau (undergraduate supervisee)
8. 2024-28 UCF Trustee Doctoral Fellowship – Ishraq Tashdid (Ph.D. supervisee)
9. 2025-26 UCF ECE Best Graduate Researcher Award – Ishraq Tashdid (Ph.D. supervisee)

Selected Talks

1. “End-to-end SoC Security Verification by Agentic AI and Formal Method (Invited)”, [2026 Secure and Private System Design for Generative & Agentic AI](#) 
2. “Toward Robust Photonic Reconfigurable Computing: Advances, Challenges, and Opportunities (Invited)”, [2026 IEEE Research and Applications of Photonics in Defense Conference \(RAPID\)](#) 
3. “Security-Centric Hardware Design in the AI Era”, Generative AI Working-group, Intel Labs, 2026 (invited).
4. “Hardware Security and Data Privacy in the Era of Emerging Technologies”, [Warren B. Nelms Annual IoT Conference 2024](#)  (invited).
5. “Navigating the Confluence: Leverage AI, Confidential Computing, and Heterogeneous Integration”, National Microelectronic Security Training Center, 2024 (invited).
6. “Hardware Design and Security in the Realm of Generative AI”, Generative AI Working-group, Intel Labs, 2024 (invited).
7. “SoC Security Tool Chain”, ACM Design Automation Conference, San Francisco, CA, 2022 (workshop).
8. Florida Institute for Cybersecurity Research (FICS) Technology Demonstration Conference (TDC), Gainesville, FL 2021 (contributed).
9. “Protecting Obfuscated Circuits against Attacks that Utilize Test Infrastructure”, FICS Research Conference, 2021 (contributed).
10. “Protecting Intellectual Property Cores against Piracy using Dynamically Obfuscated Scan Chain Architecture”, Trusted and Assured Microelectronics (TAME) Forum, Gainesville, FL, 2019 (contributed).

Funding and Grants Across Different Research Directions

1. Hardware Security:

- Sponsor: NSF RoRS; Role: Co-PI; Amount: \$299,414 (Co-PI share \$95,000); Goal: Exploring the security vulnerability of emerging neurotechnology devices; Status: [Awarded](#) 
- Sponsor: NSF Safe-OSE; Role: PI; Amount: \$1,499,877 (PI share \$950,000); Goal: Securing the OpenROAD initiative based open-source chip design ecosystem with sub-awardees - UC San Diego, and Precision Innovation Inc.; Status: Under consideration for funding.
- Sponsor: NSF; Role: PI; Amount: \$12,500 HPC; Goal: Security verification; Status: [Awarded](#) 
- Sponsor: NSF SaTC; Role: PI; Amount: \$600,000 (PI share \$300,000); Goal: Secure packaging for emerging interposers in heterogeneous integration in collaboration with LSU; Status: Under review.
- Sponsor: NSF PESOSE: Track 1; Role: Co-PI; Amount: \$300,000 (PI share \$100,000); Goal: Develop a Trust-Hub Open-Source Ecosystem for Secure Hardware Design and Verification in collaboration with University of Florida; Status: Under Review.
- Sponsor: Office of Naval Research (ONR); Role: PI; Amount: \$353,000; Goal: Security Evaluation of AI-Generated Code for Trustworthy Naval Systems; Status: Under Review.

2. AI for Chip Design: In this project we address the growing computational, reliability, and performance gap of microelectronics systems by identifying fundamental gaps

- Sponsor: NSF; Role: PI; Amount: \$38,500 HPC; Goal: Agentic AI for Hardware Design; Status: [Awarded](#) 
- Sponsor: AMD; Role: PI; Amount: 24,000 GPUH; Goal: Security verification; Status: Awarded.
- Sponsor: NSF FutureCore: SHF; Role: PI; Amount: \$600,000 (PI share \$300,000); Goal: Agentic AI for efficient, and secure hardware design - Auburn University; Status: Under Review.

Professional Services

Organizer

- 2025 IEEE World Forum on Public Safety Technology (Local Chair)
- 2024 IEEE Secure and Private Execution Environment Design (SEED) (Registration Chair)
- 2019 UF/FICS Hardware De-obfuscation Competition.

Instructor

- Micro-electronics Security Training Center.
- IEEE Young professional workshop on RTL to GDS signoff of an ASIC chip, BUET, 2017.

Technical Program Committee Member

- ACM/IEEE Design Automation Conference (DAC), 2023, 2024
- IEEE Hardware Oriented Security and Trust (HOST), 2023, 2024, 2025
- ACM Great Lakes Symposium on VLSI, 2024
- IEEE Computer Society Annual Symposium on VLSI (ISVLSI), 2024
- IEEE International Conference on Computer Design (ICCD)

Reviewer

- IEEE Transaction on Computer Aided Design (TCAD)
- IEEE Transactions on Computers
- IEEE VLSI Test Symposium (VTS)
- IEEE International Test Conference (ITC)
- ACM Transactions on Design Automation of Electronic Systems (TODAES)
- ACM Hardware and System Security (HaSS)
- Springer Nature Computer Science
- ACM Journal on Emerging Technologies in Computing Systems (JETC)
- ACM/IEEE Design Automation Conference (DAC)
- IEEE Hardware Oriented Security and Trust (HOST)
- ACM Great Lakes Symposium on VLSI (GLSVLSI)
- IEEE Computer Society Annual Symposium on VLSI (ISVLSI)
- IEEE International Conference on Computer Design (ICCD)